



# Integrated HPLC–UV validation and LC–MS/MS optimization with molecular networking and chemometrics for advanced characterization of chlorogenic acids in *Ilex guayusa*

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## ABSTRACT

Chlorogenic acids (caffeoylquinic acid isomers, CQAs) are major phenolic constituents of *Ilex guayusa*, but their comprehensive profiling in complex plant matrices is hindered by co-elution, overlapping UV spectra, and isomeric similarity in MS/MS. Rather than aiming to fully resolve isomer-specific quantification by MS, here we present an integrated workflow that couples validated HPLC–UV quantification of the major CQA (5-CQA) with an optimized UPLC–MS/MS strategy designed to improve MS1 peak integrity and expand MS/MS coverage for higher-confidence structural annotation. The HPLC–UV method showed excellent performance for targeted quantification of 5-CQA, including strong linearity ( $r^2 = 0.998$ ), selectivity, sensitivity (LOQ = 0.25 mg/L), precision, and recovery. For LC–MS/MS, FastDDA acquisition (top-5 vs. top-15 precursors) revealed the expected trade-off between fragmentation depth and MS1 peak quality; however, post-acquisition raw-data merging restored MS1 fidelity and increased the number of detected features by 43%, enabling high-confidence annotation rather than quantitative discrimination of 16 metabolites and the propagation of oxidized CQA-related derivatives using feature-based molecular networking. Multivariate analyses (PCA, volcano plots, HCA) indicated that geographic location exerted the strongest influence on the metabolite composition, followed by sunlight exposure and plant age. Overall, the proposed workflow provides a practical framework that integrates robust chromatographic quantification with MS acquisition and data-processing optimization, thereby enhancing structural characterization and biological interpretation, rather than complete isomer-resolved quantification, of chlorogenic-acid-related chemistry across complex plant-derived and natural product matrices

## 1. Introduction

*Ilex guayusa* (Loes.), commonly known as guayusa, is a native plant of the Amazonia, distributed mainly in Colombia, Peru, and Ecuador. In Ecuador, the Napo province represents one of its principal centres of production and consumption [1]. The use of guayusa as an infusion is an

ancestral practice among indigenous communities of eastern Ecuador, who attribute to it both stimulating and spiritual properties, integrating the plant into their cultural traditions and rituals [2].

Guayusa is an evergreen tree that can reach up to 30 meters in height, with oval to elliptical leaves measuring 7–22 cm in length, which are traditionally used for infusions [1]. Unlike other *Ilex* species, no

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subspecies or botanical varieties of *I. guayusa* have been described to date. Its cultivation is strongly associated with traditional agroforestry systems known as “chakras,” managed by indigenous communities who integrate agricultural production with biodiversity conservation [3]. These systems promote organic agricultural practices, free of synthetic fertilizers and pesticides, thereby supporting both product and ecosystem sustainability [4].

From a phytochemical standpoint, *I. guayusa* shares similarities with Yerba maté (*Ilex paraguariensis*), as it contains methylxanthines such as caffeine and theobromine, together with phenolic compounds with well-documented antioxidant activity [5,6]. Among these, caffeoylquinic acid (CQA) isomers have drawn significant attention due to their antioxidant and anti-inflammatory properties, as well as their potential roles in reducing visceral fat and improving lipid metabolism [7]. These isomers constitute a family of esters derived from quinic acid and cinnamic acids such as caffeic, ferulic, and p-coumaric acids, playing a key role in regulating oxidative stress in plants and offering promising applications for the nutraceutical industry [8]. Despite their biological importance, the environmental and physiological determinants influencing CQA isomer accumulation in *I. guayusa* have not been systematically investigated.

From an analytical perspective, resolving and confidently annotating CQA isomers presents significant analytical challenges, particularly due to co-elution, overlapping UV spectra, and isomeric similarity in MS/MS fragmentation. Conventional HPLC–UV-vis methods remain essential for robust quantification, particularly of major isomers such as 5-CQA, but their capacity to distinguish structurally similar positional isomers is limited due to overlapping UV spectra and co-eluting chromatographic behavior. Thus, when the scope extends to comprehensive profiling of multiple CQA isomers and their oxidized derivatives, LC–MS/MS becomes indispensable. However, untargeted identification of chlorogenic isomers introduces additional complexity: precursor ions may share nearly identical *m/z* values and retention times, and the richness of fragmentation data strongly depends on acquisition parameters and data-processing strategies.

Recent strategies such as FastDDA aim to expand MS/MS coverage by increasing the number of precursor ions selected per cycle, though excessive selection can reduce MS1 scan density, compromise chromatographic peak shape, and hinder feature detection. Optimizing this balance, together with post-acquisition raw-data merging, has proven effective for improving MS1 peak integrity and enhancing fragmentation depth, thereby supporting more reliable metabolite annotation [9]. However, these strategies primarily improve data quality and annotation confidence rather than providing full isomer-resolved quantitative discrimination. Additionally, complementary tools such as molecular networking assist in organizing MS/MS data into structurally meaningful clusters, facilitating the annotation of low-abundance derivatives and improving compound identification through the recognition of similar fragmentation patterns. Despite these analytical advancements, no study has yet integrated validated HPLC–UV quantification, optimized FastDDA-based LC–MS/MS acquisition, raw-data merging, molecular networking, and chemometric analysis to characterize CQA isomers in *I. guayusa*. Recent literature, including systematic and bibliometric reviews, has extensively summarized the phytochemical composition, biological activities, and application trends of *Ilex guayusa* [1]. However, these studies largely adopt descriptive perspectives and do not address the analytical challenges associated with the accurate quantification, structural annotation, and isomeric complexity of chlorogenic acids. The present work addresses this gap by introducing an experimentally validated analytical workflow focused on chromatographic robustness and structure-oriented MS data interpretation.

Environmental variables, particularly leaf maturity, sunlight exposure, and geographic location, are known to modulate secondary metabolite biosynthesis in *Ilex* species [10–13]. Understanding how these factors influence the CQA profile of *I. guayusa* requires an analytical framework capable of coupling reliable quantification with

confident structural characterization. Accordingly, this study is structured around an integrated workflow in which validated HPLC–UV quantification provides a robust quantitative anchor, while optimized LC–MS/MS acquisition, data-processing strategies, molecular networking, and chemometric analysis enable comprehensive structural annotation and interpretation of environmentally driven variation. Using a factorial mixed-level experimental design, this approach provides a transferable framework for the characterization of chlorogenic-acid-related chemistry in complex plant-derived and natural product matrices, without aiming at full isomer-resolved MS quantification.

## 2. Materials and methods

The Materials and methods section is organized to reflect the integrated nature of the proposed workflow, in which chromatographic quantification, MS acquisition and data-processing optimization, molecular networking, and chemometric analyses are implemented as interdependent components of a single analytical system.

### 2.1. Experimental design

A mixed level factorial experimental design was implemented, structured in three blocks factors: location (three levels), leaf maturity (three levels), and sunlight exposure (two levels) (Table 1). The locations correspond to three traditional agroforestry systems (“chakras”): Chakra A (Talag), Chakra B (Alto Pano), and Chakra C (Alto Tena), following the geographic categorization proposed by Dunn, et al. (2018) [14]. Leaf maturity was defined according to plant age, with three levels: Early (0), corresponding to plants aged 4–6 years; Medium (1), corresponding to plants aged 6–8 years; and Late (2), corresponding to plants aged 8–10 years, based on evidence indicating developmental changes in phytochemical profiles [15]. Sunlight exposure was classified into Light (+), for leaves collected from plants exposed to  $\geq 300$  PPFD ( $\mu\text{mol}/\text{m}^2/\text{s}$ ), and Shade (–), for plants receiving 0–200 PPFD. Light intensity was measured *in situ* using a lux meter according to Liza G. (2018) [16].

Leaf sampling was carried out using a randomized collection procedure adapted from Craig A. et al. (2016) [17], Farmers affiliated with the Kallari guayusa trade network provided information regarding plant age and traditional harvesting criteria. Following ancestral practices, leaves with small reddish tips (“gogoyos”) from early branches were avoided; instead, mature dark-green leaves, traditionally selected for infusion, were prioritized. For each combination of factors (location  $\times$  maturity  $\times$  light exposure), leaves were collected, placed in labeled paper bags, transported at 4°C to the laboratory, washed with sterile water, and stored at –80°C until analysis.

### 2.2. High performance liquid chromatography (HPLC–UV–Vis) analysis

#### 2.2.1. Sample preparation

Samples (leaf material collected from individual plants under different conditions) were prepared following a modified protocol from Craig A. et al. (2016) [17]. Fresh guayusa leaves (15 mg) were weighed into a 50 mL volumetric flask, to which 40 mL of diluent (aqueous

**Table 1**  
Experimental design showing factors and levels.

|                  |            | Chakra A  |           | Chakra B  |           | Chakra C  |           |
|------------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Light conditions |            | Light (+) | Shade (–) | Light (+) | Shade (–) | Light (+) | Shade (–) |
| Age              | Early (0)  | A0+       | A0–       | B0+       | B0–       | C0+       | C0–       |
|                  | Medium (1) | A1+       | A1–       | B1+       | B1–       | C1+       | C1–       |
|                  | Late(2)    | A2+       | A2–       | B2+       | B2–       | C2+       | C2–       |

solution with 2.0% formic acid and 10% acetonitrile) was added. The mixture was sonicated for 20 min with brief pauses every 5 min, followed by manual shaking to ensure homogenization. The final volume was adjusted to the mark of the volumetric flask. The homogenized extract was then filtered through 0.45 µm PTFE syringe filters, and the filtrate was transferred into 1.5 mL vials for analysis.

## 2.2.2. System conditions

The analysis was performed on a Waters High-Performance Liquid Chromatography (HPLC) system equipped with a 1525 binary pump, 2489 UV/Visible detector, and 2707 autosampler. Chromatographic separation was carried out using a Phenomenex Gemini C18 column (150 × 2 mm, 5 µm particle size). The mobile phase consisted of solvent A (0.1% formic acid in water) and solvent B (HPLC-grade acetonitrile with 0.1% formic acid), delivered at a flow rate of 0.75 mL/min. The gradient program was as follows; 5% B (10 min), 5–20% B (5 min), 20–5% B (0.1 min), 5–95% B (1.4 min), followed by an isocratic hold at 5% B. The injection volume was 10 µL, and detection was performed at 330 nm. A calibration curve was prepared from a 100 mg/L stock solution of 5-CQA (Chlorogenic acid, Supelco Pharmaceutical Secondary Standard; Certified Reference Material), following methods adapted from Craig et al. (2016) [17]. The working range for quantification was 1.25–25 mg/L.

## 2.2.3. Analytical method development process

For the analytical method validation, the parameters evaluated included selectivity, limits of detection and quantification (LOD/LOQ), repeatability, and reproducibility following the Eurachem Guide (2005) [18,19].

The acceptance criteria were defined as follows: linearity with  $r^2 \geq 0.99$ ; selectivity demonstrated by the absence of signal in the reagent blank and the presence of signal in the matrix blank; LOD corresponding to  $\leq 50\%$  recovery percentage; LOQ corresponding to 90%–110% recovery; precision with %CV  $\leq 12\%$  (repeatability) and  $\leq 15\%$  (intermediate precision); and veracity expressed as recovery between 90%–110%.

**2.2.3.1. Linearity.** Linearity was evaluated using five independent calibration curves for 5-CQA (chlorogenic acid), followed by statistical analysis. The linear correlation coefficient ( $r^2$ ) was used as the primary acceptance criterion. The percentage of linearity was calculated from the calibration curve  $r^2$  and its standard deviation, according to the following Eq. (1):

$$\%Linearity = \left( \frac{1 - S_b}{b} \right) * 100 \quad (1)$$

Where  $S_b$  is the standard deviation of the slope  
 $b$  is the slope

**2.2.3.2. Selectivity.** Selectivity was assessed using two approaches: a reagent blank, analyzing the diluent used for sample and standard preparation, and a matrix blank, analyzing an aromatic plant sample containing 5-CQA.

**2.2.3.3. Limit of detection.** The Limit of detection was determined by preparing 10 replicates of 5-CQA standard solutions below the lowest point of the working range (0.025, 0.05, 0.1, 0.2, 0.25, 0.5, and 1 mg/L). The LOD was defined as the concentration of the lowest-level solution that met the proposed verification criteria ( $\leq 50\%$  recovery), according to the following Eq. (2):

$$LOD = \frac{3 * S_b}{b} \quad (2)$$

Where  $S_b$  is the standard deviation of the slope  
 $b$  is the slope

**2.2.3.4. Limit of quantification.** The limit of quantification was determined by spiking a sample of known 5-CQA concentration with additional 5-CQA at levels below the working range (0.025–1 mg/L). Each concentration was analyzed in 10 replicates, and the LOQ was defined as the lowest concentration that met the verification criteria (90%–110% recovery), according to the following Eq. (3):

$$LOQ = \frac{10 * S_b}{b} \quad (3)$$

Where  $S_b$  is the standard deviation of the slope  
 $b$  is the slope

**2.2.3.5. Precision.** Three concentration levels of the analyte were selected: high (25 mg/L), medium (3.6 mg/L), and low (1.75 mg/L). For each level, four independent replicates were prepared under identical conditions to ensure consistency in sample preparation and handling.

Repeatability (intra-day precision) was assessed by analyzing each replicate three times within the same day. This procedure was repeated over three consecutive days under identical experimental conditions to evaluate intermediate precision (inter-day variation).

The coefficient of variation (%CV) was calculated for each concentration level and each day based on the triplicate measurements to estimate repeatability. Subsequently, daily mean values were used to calculate the %CV across the three days, providing a measure of intermediate precision. All %CV values were compared with the predefined acceptance criteria of  $\leq 12\%$  for repeatability and  $\leq 15\%$  for intermediate precision, as recommended by Eurachem (2005) [18], according to the following Eq. (4):

$$\%CV = \left( \frac{SD}{mean} \right) * 100 \quad (4)$$

Where SD is the standard deviation

**2.2.3.6. Veracity.** At least two concentration levels were selected for sample spiking: low (0.4 mg/g) and high (3.6 mg/g), both within the method's working range. A total of twelve determinations were carried out, and recovery percentage was evaluated against predefined verification. Spiked samples (with known amounts of analyte added to the matrix) were analyzed using the method under evaluation. Recovery percentages values were calculated after converting results from mg/L to mg/g.

Finally, method performance was assessed by verifying whether recovery percentages fell within the acceptable range. The entire method verification process was conducted in accordance with Eurachem (2005) [18]. Control chart scripts were made publicly through the GitHub repository ([https://github.com/IKIAM-NPLab/I\\_guayusa\\_caffeoylquinic\\_acid\\_isomers](https://github.com/IKIAM-NPLab/I_guayusa_caffeoylquinic_acid_isomers)).

## 2.3. Statistical analysis

A Generalized Linear Model (GLM) was fitted using the Gaussian family with a log link function to describe the relationship between the independent variables and the transformed response variable [20]. Model diagnostics were performed to evaluate the assumptions of linearity, independence, homoscedasticity, and normality of residuals using scatterplots, residuals versus fitted values plots, and Q-Q plots. Subsequently, an analysis of variance (ANOVA,  $p < 0.05$ ) was carried out to assess the influence of categorical variables and their interactions on the response variable. Post hoc comparisons among predictor categories were conducted using Tukey's HSD test. The complete statistical analysis scripts are openly available in the GitHub repository ([https://github.com/IKIAM-NPLab/I\\_guayusa\\_caffeoylquinic\\_acid\\_isomers](https://github.com/IKIAM-NPLab/I_guayusa_caffeoylquinic_acid_isomers)).

## 2.4. LC-MS metabolomic profile

### 2.4.1. Sample preparation

Samples were prepared following the HPLC sample preparation protocol (Section 2.2.1), the 5-CQA concentration determined by HPLC-UV-Vis was used to dilute the samples to a final concentration of 1 000 mg/mL. Subsequently, extraction blanks (ultrapure water), Quality Control samples (QC, pooled from all samples), subQC (pooled by factor: location, sunlight exposure and leaf maturity) and individual samples were prepared.

### 2.4.2. System conditions

The metabolomic profiling focused on caffeoylquinic acid isomers was carried out using a Waters Acquity I-Class LC system coupled to a XEVO G2-XS QTOF mass spectrometer. Samples (Waters Corporation, Milford, MA, USA). Samples, blanks, QC, and subQC were analysed with an injection volume of 3  $\mu$ L. Chromatographic separation was achieved on an ACQUITY UPLC<sup>®</sup> CSH<sup>™</sup> C18 1.7  $\mu$ m (3.0  $\times$  50 mm) equipped with an Acquity UPLC CSH C18 VanGuard precolumn (2.1  $\times$  5.0 mm). Reversed-phase chromatography was performed using water with 0.1% formic acid (mobile phase A) and acetonitrile with 0.1% formic acid (mobile phase B) at a flow rate of 0.5 mL/min. The elution program was as follows: 0–0.5 min 1% B; 0.5–8 min, linear increase to 100 % B; 8–9 min hold 100 % B; 9.01–10.0 min, re-equilibration at 1 % B. Full-scan mass spectra were acquired in both positive and negative electrospray ionization (ESI) modes to ensure comprehensive metabolite coverage. Optimized MS parameters included a capillary voltage of 2.5 kV (negative mode) and 0.5 kV (positive mode), source temperature of 100°C, desolvation temperature of 500°C, nitrogen desolvation gas at 1 000 L/h, cone gas at 0 L/h, and argon as the collision gas. The acquisition mass range was set from 100–1200  $m/z$  for full MS scans. Data were mass-calibrated using a continuous infusion of leucine–enkephalin (1 ng/ $\mu$ L) at 10  $\mu$ L/min as an external LockMass reference.

Using the same MS settings, MS/MS spectra of the pooled QC samples were acquired with a FastDDA method in both positive and negative ionization modes at collision energies of 10, 20, 40 and 60 eV. Data acquisition and processing were performed using MassLynk software (Waters Corporation)

MS/MS acquisition parameters included a mass range of 100–1200  $m/z$  for MS scans (survey scan time of 0.2 s), and 50–1200  $m/z$  for MS/MS scans (survey scan time: 0.1 s). Two acquisition modes were employed: FastDDA 5 (selection of the five most intense precursor ions per MS scan) and FastDDA 15 (selections of the fifteen most intense precursor ions). Collision-induced dissociation (CID) was applied at energies of 10, 20, 40 and 60 eV. Automatic MS/MS acquisition was triggered when the total ion current (TIC) intensity exceeded 2500 s<sup>-1</sup> and reverted to the MS scanning after one MS/MS event. Peak detection was performed using the “Include by Charge State” option with default parameters for a singly charged ions, applying a 2 s exclusion window to avoid repeated fragmentation of the same precursor ions.

### 2.4.3. Data processing for future metabolomic identification

We use the term “function” to refer to a specific data channel in vendor chromatograms (\*.raw files); Function 1 corresponds to MS level 1 (MS or MS1), also known as the full scan, whereas the subsequent functions correspond to MS level 2 (MS2 or MS/MS). In the case of FastDDA, the number of functions is equal to the number of precursor ions fragmented per MS1 scan (in our case five functions for FastDDA 5, and fifteen functions for FastDDA 15). The function following the MS/MS scans corresponds to the lockmass. We used ProteoWizard MSConvert (version 3.0.24155-80747de) [21] to convert \*.raw files to \*.mzML format, and the raw data were deposited in GNPS/MassIVE (MSV000098557). Before converting the \*.raw files to \*.mzML, we used Water2mzml (AnP311/Waters2mzML) to remove the leucine lockMass signal from both full-scan and MS/MS data as we explained in Supplementary tutorial information. The filters applied in MSConvert included

“peak picking” using the vendor algorithm and “subset”. For the FastDDA datasets (FastDDA 5 and FastDDA 15), two approaches were implemented to improve downstream data analysis: In the first approach, the raw FastDDA data (MS, MS/MS) were split into separate \*.mzML files, one containing the MS function and another containing the MS/MS functions, according to the Waters user manual. Additionally, for this approach, the full-scan MS data from the corresponding sample, as well as from the QC pool sample, were converted to \*.mzML format and subsequently merged with the previously separated MS/MS data from the FastDDA QC pool sample using MZmine 4.5.20 and the “Merge raw data files” algorithm [22]. In the second approach, the raw FastDDA data was converted directly into \*.mzML files without splitting. As shown in the results section, the second approach improved the chromatographic peak shape of the FastDDA data, thereby allowing the extracting of more MS/MS information. In summary, two groups of data were analyzed: one group containing FastDDA MS/MS data merged with full-scan MS data, and another group containing the raw FastDDA data.

The resulting files were processed in MZmine 4.5.20 [22] to identify optimal parameters for peak deconvolution, following the non-targeted metabolomics workflow [23]. The processing pipeline included mass detection, chromatogram builder (ADAP), smoothing, local minimum feature resolved, isotope grouping, aligned and filtering. Detailed parameters are available in **Supplementary Tables S1.1–S4.2**. Finally, MZmine exported two key outputs: (i) a .csv feature height table \*containing precursor ion intensity data, and (ii) a \*.mgf file containing MS/MS fragment  $m/z$  and intensity values corresponding to the detected features

**2.4.3.1. Level 1 Metabolite Identification.** For metabolomic identification, the analysis was conducted in accordance with the minimum standards proposed by the Metabolomics Standards Initiative (MSI) [24]. Level 1 identification was assigned for compounds detected as reference standards within QC samples, where MS1 features were required to align with a retention time (RT) difference of  $\leq \pm 0.05$  min and a mass error of  $\leq \pm 10$  ppm **Supplementary Information Fig. S6**). Additionally, for compounds of particular interest (5-CQA and 3-CQA), the MS/MS spectra were visually inspected to confirm the alignment of the most abundant fragment ions between the reference standard and the experimental spectra.

**2.4.3.2. Level 2 Metabolite Identification.** Level 2 identification was based on MS/MS data and the experimental precursor mass, which were compared against public spectral databases through the GNPS platform [25] using the Feature-Based Molecular Networking (FBMN) workflow [25,26]. Analyses were performed under different collision energies, ionization polarities and Fast DDA acquisition modes. The FBMS parameters included a fragment ion mass tolerance of 0.02 Da, a minimum of four aligned fragment ions, and a cosine similarity score  $\geq 0.7$ . For each putatively identified metabolite, the mass accuracy (ppm error) was calculated as the difference between the experimental mass (measured by the instrument) and the theoretical monoisotopic mass. Only metabolites meeting both criteria, mass error  $\leq \pm 10$  ppm and cosine score  $\geq 0.7$ , were retained for downstream analysis

### 2.4.4. Molecular network analysis

The MS/MS data deconvoluted in MZmine were transferred to GNPS using the Feature-Based Molecular Networking (FBMN) workflow in both negative (ID=9e20d56859844541bd4a2a37d95e08c3) and positive ionization modes (ID=2f18f1c6dc1349c385b2 fc41ae32ddb0), according to the GNPS documentation [27]. Network node connection parameters included a cosine score  $\geq 0.7$ , a minimum of six aligned fragments ions, and a maximum precursor mass shift of 500 Da. Subsequently, the “merge polarity” algorithm was applied to combine the positive and negative ionization networks into a polarity-merge molecular network (ID=69e318677afc4057bdab12927a18fcfd). The

merging parameters were set to 50 ppm mass tolerance and 1 min of retention time tolerance for aligning features between runs. Network nodes were annotated by comparing experimental spectra against GNPS spectral libraries using the same parameters as those applied for Level 2 identification.

Additionally, several complementary tools were used to propagate and annotate compounds within the networks, including Dereplicator+, MS2LDA, MS-FINDER, and SIRIUS. For Dereplicator+ (positive polarity: ID=687d8d57c6644c22baa076d52be26263; negative polarity: ID=b108c0b7944547eabc2b0643018424a9), parameters included a precursor ion mass tolerance of 0.005 Da and a fragment ion mass tolerance of 0.01 Da.

For MS2LDA (positive polarity: ID=a4af075cb207488882d57f69c27106d0; negative polarity: ID=a23a6f0a2ad445eeb697ff44ecd3a41d), parameters included an overlap score threshold of 0.3, a probability value threshold of 0.1, and topX nodes = 5, with all motif databases enabled. For compound annotations in MS-FINDER (v3.61), the option “formula prediction and *in silico* structure elucidation from fragmentation” was used with the following libraries: Internal Experimental Library, Food, Plant Biomolecules, and Natural Products MS/MS Libraries. SIRIUS (v6.1) was configured with instrument type = Q-TOF, mass accuracy = 10 ppm, and possible adducts [M-H]<sup>−</sup> (negative polarity) and [M+H]<sup>+</sup> (positive polarity). CSI:FingerID was employed for structure elucidation, with all databases enabled except DSSTox. Extended parameters for MS-FINDER and SIRIUS are available in the (Supplementary Tables S6 and S7). Finally, the resulting molecular networks were visualized using Cytoscape [28].

#### 2.4.5. Multivariate metabolomic statistics

The full-scan data used for multivariate metabolomic analyses were converted from vendor raw files (\*.raw) to NetCDF (\*.CDF) format using the DataBridge conversion utility included in MassLynx software 4.1v (Waters Corporation). This program generates individual \*.CDF files according to the number of functions present; each file contains the information corresponding to a specific function. For this analysis, we used the first \*.CDF file, which contains the information from Function 1 (full scan). The data were processed following the same MZmine workflow used for metabolite identification, with some modifications (Supplementary Tables S9–S10). The processing pipeline included mass detection, chromatogram building (ADAP), smoothing, local minimum resolver, isotope grouping, alignment, gap filling, duplicate filtering, and precursor mass searching against a local database to annotate metabolites at MSI levels 1 and 2.

Finally, MZmine exported a feature intensity table (\*.csv format) containing precursor ion intensities and annotated metabolites. This dataset was imported into RStudio (version 2024.12.1+563) for multivariate statistical analyses using the “notame” package [29]. Multivariate statistical analyses were performed to explore metabolomic variation associated with each experimental factor independently. Because each factor represented a different type of biological or environmental contrast, specific statistical approaches were applied accordingly: (i) Principal Component Analysis (PCA) was used to assess the effect of geographical location, as this unsupervised method efficiently summarizes variance among samples and allows visualization of spatial clustering patterns. (ii) A Volcano plot analysis was performed to evaluate sunlight exposure, which represents a binary environmental contrast; this approach enables direct visualization of metabolites showing both statistical significance and fold-change differences between sunlight exposure. (iii) A Heatmap combined with Hierarchical Cluster Analysis (HCA) was used to explore age-related metabolic trends, allowing the comparison of relative metabolite abundances and clustering of age groups based on similar metabolic profiles. This integrative approach enabled the identification of both global and factor-specific metabolic differences. All The analysis scripts were uploaded to the GitHub repository ([https://github.com/IKIAM-NPLab/Iguayusa\\_caffeoylquinic\\_acid\\_isomers](https://github.com/IKIAM-NPLab/Iguayusa_caffeoylquinic_acid_isomers)).

### 3. Results and discussion

The Results and discussion are presented following the logic of the integrated workflow, showing how each analytical component builds upon the previous one to collectively support quantification, structural annotation, and biological interpretation.

#### 3.1. Method verification

Table 2 summarizes the main validation parameters of the HPLC–UV method for 5-caffeoylquinic acid (5-CQA), including linearity, sensitivity, precision, and recovery, in accordance with the Eurachem guidelines.

##### 3.1.1. Linearity

The calibration curves exhibited an average correlation coefficients of  $r^2 = 0.998$ , meeting the acceptance criterion for method verification ( $r^2 \geq 0.99$ ). Such linearity ensures reliable quantification of 5-CQA within the specific concentration range. These results are consistent with literature recommendations, as outlined in the Eurachem Guide: The Fitness for Purpose of Analytical Methods [18].

##### 3.1.2. Selectivity

No peaks were observed in the reactive blank chromatogram (Supplementary Information Fig. S1) at the retention times corresponding to 5-CQA (12 min). However, 5-CQA is naturally present in most aromatic plants [30–32]. To assess selectivity, the analyte was evaluated in the matrix blank in accordance with Eurachem Guide (2016) [18], which defines selectivity as the method’s ability to reliably detect the target analyte in presence of other components. Selectivity was verified by detecting 5-CQA across multiple infusion leaf species, using chamomile (*Chamaemelum nobile*) and coriander (*Coriandrum sativum*), prepared under the same conditions (Supplementary Information Fig. S2.1 and S2.2), identification was supported by previous studies for chamomile [33] and coriander [34].

The absence of peaks in the reactive blank confirms that no interfering metabolites are present from sample processing, while the matrix blank demonstrates that the detected signal corresponds specifically to 5-CQA, which is naturally found in most plants used for infusions.

##### 3.1.3. Limit of detection and limit of quantification

The LOD and LOQ were 0.05 mg/L and 0.25 mg/L, respectively. These values were determined by analyzing ten replicates at concentrations below the lowest point of the calibration curve (1, 0.5, 0.25, 0.2, 0.1, 0.05 mg/L). The LOQ was defined as the lowest concentration

**Table 2**  
Summary of HPLC–UV method validation parameters for 5-caffeoylquinic acid (5-CQA).

| Parameter                                      | Specification                                                                           | Result                                                                                              |
|------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Linearity                                      | $r^2 \geq 0.99$                                                                         | 0.998                                                                                               |
| Selectivity                                    | Absence of signal in reagent blank at analyte RT and presence of signal in matrix blank | Signal observed at the retention time of 5-CQA in matrix; no interference detected in reagent blank |
| Limit of detection (LOD)                       | Recovery $\leq 50$ %                                                                    | 0.05 mg L <sup>−1</sup>                                                                             |
| Limit of quantification (LOQ)                  | Recovery 90–110 %                                                                       | 0.25 mg L <sup>−1</sup>                                                                             |
| Precision – Repeatability (intra-day)          | %CV $\leq 12$ %                                                                         | High: 0.01–5.65 %<br>Medium: 4.32–8.18 %<br>Low: 5.94–40.02 %                                       |
| Precision – Intermediate precision (inter-day) | %CV $\leq 15$ %                                                                         | High: 7.59 %<br>Medium: 3.94 %<br>Low: 11.30 %                                                      |
| Accuracy (Recovery)                            | 90–110 %                                                                                | 95–110 %                                                                                            |

exhibiting acceptable precision, with recovery percentage between 90 and 110%, within the linear response range, as shown in Supplementary information Fig. S3. The LOD corresponds to the minimum concentration detectable by the instrument, characterized by a recovery percentage  $\leq 50\%$ . These parameters are essential for evaluating the method's sensitivity [35], demonstrate that the procedure fulfills the required sensitivity criteria for the reliable detection and quantification of chlorogenic acid in the analyzed samples [18].

### 3.1.4. Precision

The evaluation of repeatability (intra-day %CV) and intermediate precision (inter-day %CV) is presented in **Supplementary information Fig. S4** and **Supplementary information Table S4.1**, which report the mean concentration and coefficient of variation (%CV) at three distinct concentration levels (low, medium, high) over three consecutive days. This assessment aimed to verify the robustness and reliability of the analytical method under routine laboratory conditions.

For intermediate precision, reflecting variability in daily mean concentrations across three days, %CV values were 3.94% for the medium level, 7.59% for the high level, and 11.30% for the low level. All values were well below the general acceptance threshold of 15% for intermediate precision, confirming that the method maintains consistent performance over time despite potential fluctuations in environmental conditions or daily laboratory practice.

Regarding repeatability, %CV values ranged between 4.32% to 8.18% for the medium level, 0.01% to 5.65% for the high level, and 5.94% to 40.02% for the low level. These values represent intra-day variability observed when analyzing the same concentration multiple times within a single day. Both medium and high concentrations remained within the accepted limit for repeatability (%CV  $\leq 12\%$ ) as recommended by Eurchem [18]. However, the low concentration level exceeded this limit on the second and third days (21.46% and 40.02%, respectively), indicating greater intra-day variability at this level. Such behavior is commonly observed near the LOQ and may be exacerbated by 5-CQA volatilization, leading to inconsistencies among replicate measurements [35].

Overall, these results demonstrate that the method is sufficiently precise and robust for quantifying the analyte at medium and high concentrations, both within and across days. Nevertheless, caution is advised when interpreting results at low concentrations, where higher intra-day variability may affect quantitative reliability.

### 3.1.5. Veracity

The recovery percentages for the low (0.4 mg/g) and high (3.6 mg/g) concentrations levels were within the accepted range of 90 to 110 % [18]. The method's veracity was further confirmed by recoveries within this acceptable range, as illustrated in Supplementary information Fig. S5.

The combined evaluation of linearity, selectivity, LOD, LOQ, precision, and veracity demonstrates the reliability and suitability for the quantification of 5-CQA.

## 3.2. Quantification analysis of chlorogenic acid (5-CQA) by high performance liquid chromatography HPLC-UV-Vis

The analysis revealed significant variations in 5-CQA concentrations associated with environmental and cultivation factors (Fig. 1; **Supplementary Table E1**). The model assumption checks indicated that the residuals displayed an appropriate distribution (KS = 0.121,  $p = 0.407$ ), without heteroscedasticity (testDispersion = 1.215,  $p = 0.408$ ) or outlier influence, and with low multicollinearity among predictors (all VIF  $< 5$ ).

These outcomes confirmed that the data exhibited a normal and homogeneous dispersion under the GLM framework detailed in the project repository ([https://github.com/IKIAM-NPLab/Iguayusa\\_caffeoylquinic\\_acid\\_isomers/blob/main/Notebooks/Statistical\\_factory\\_design.md](https://github.com/IKIAM-NPLab/Iguayusa_caffeoylquinic_acid_isomers/blob/main/Notebooks/Statistical_factory_design.md)). Accordingly, the ANOVA results indicated statistically

significant effects of chakra (collection site) and sunlight exposure, whereas plant age showed a marginal effect ( $p = 0.0678$ ). Although this factor did not reach conventional significance at  $\alpha = 0.05$ , Tukey's HSD test indicated consistent pairwise differences among age groups, supporting a biological trend in the data. These differences were further reinforced by the observed interaction between age and sunlight exposure, suggesting that environmental conditions modulate age-dependent metabolic responses.

The collection site (chakra) had a major influence on 5-CQA concentration, likely due to differences in soil quality or other environmental variables [36]. As previously reported [34], where a higher quality of soil is observed for Alto Pano, while Alto Tena presents elevated Zn and Cu levels and Talag exhibits nutrient depletion. These soil characteristics are consistent with the observed spatial variations in 5-CQA concentration among locations.

Plant age also appeared to play a relevant role in 5-CQA accumulation: early-stage plants (4–6 years) exhibited higher concentrations (**Supplementary Table E1**), likely because younger individuals produce greater amounts of phenolic antioxidants to regulate growth and maintain metabolic activity [37–39]. Sunlight exposure appeared to modulate this pattern (Fig. 1B). Although the ANOVA indicated a statistically significant overall effect of light exposure ( $p = 0.0004$ ), Tukey's post-hoc test did not reveal significant pairwise differences between light and shaded conditions. This apparent discrepancy suggests that the observed significance may arise primarily from the strong interaction between age and light ( $p = 2.20 \times 10^{-16}$ ), rather than from the main effect of light alone. Descriptively, 5-CQA concentrations tended to be higher in sunlight young plants but declined with age, whereas shaded plants maintained relatively stable levels throughout development stages (6–10 years). This trend suggests that light and age jointly influence 5-CQA biosynthesis [38], with shaded conditions appearing to preserve 5-CQA content more effectively, regardless of plant age.

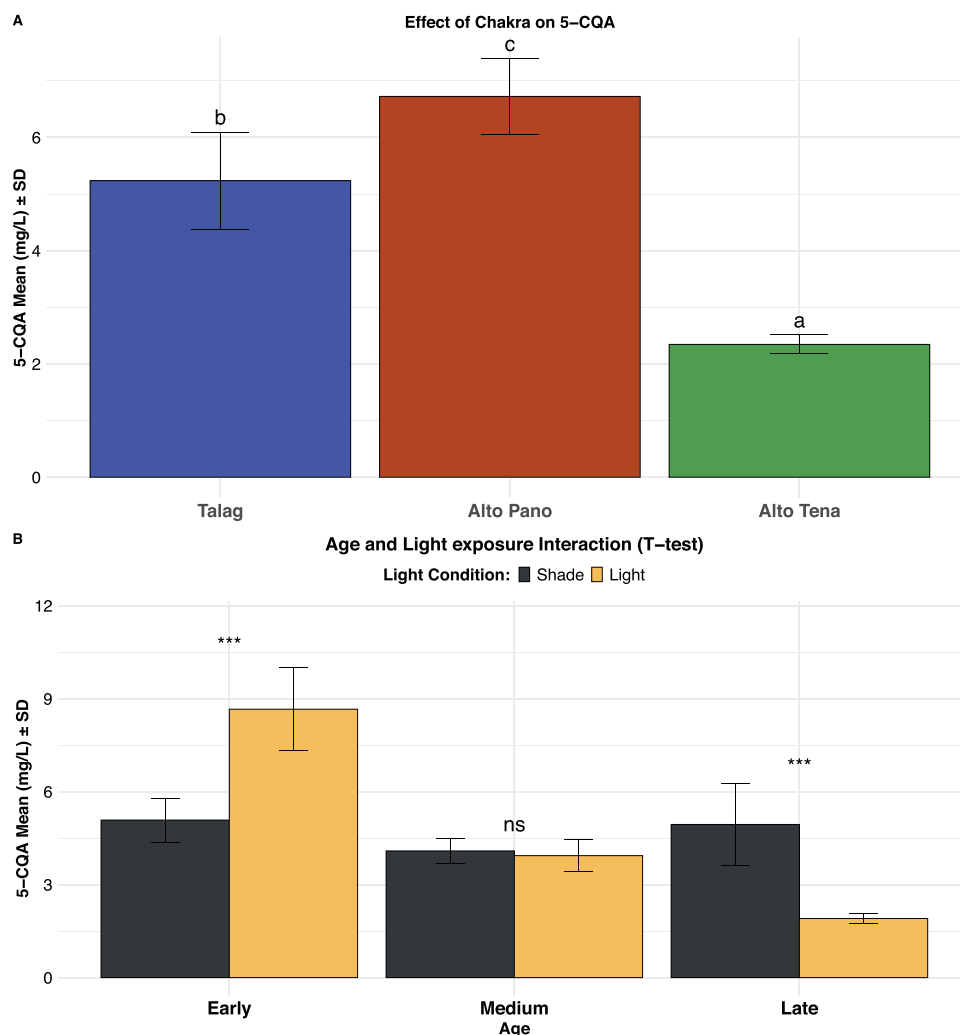
Overall, the most favorable conditions for achieving elevated 5-CQA concentrations in *Ilex guayusa* leaves were observed in early-stage (4–6 years) grown under sunlight exposure. Nevertheless, as plants age, those cultivated under shade retain comparatively higher 5-CQA levels. Therefore, for harvesting strategies targeting mature plants ( $>6$  years), shade-grown guayusa represents a more suitable alternative.

## 3.3. Analysis of caffeoylquinic acid isomers profile by UPLC-MS/MS

In *Ilex guayusa*, a total of 16 metabolites were identified (Table 3). Although the number of detected metabolites appears limited, the sample treatment was specifically optimized to favor the extraction of chlorogenic acids.

The metabolite identification was performed using MS/MS scores, with cosine similarity evaluated through the GNPS platform. A quinic acid metabolite was identified in negative ionization mode ( $m/z$  rt) 191.0555\_0.5889 min  $[M-H]^-$  and assigned level 2 of identification; this metabolite serves as a precursor for caffeoylquinic acids [39]. Several caffeoylquinic acid isomers were identified as level 1 using authentic standards, including neochlorogenic acid and chlorogenic acid. In addition, caffeine was detected in positive ionization mode 195.0885\_2.4233 min  $[M+H]^+$  at level 1 represents the most abundant metabolite in guayusa leaves samples. Several biologically active flavonoids were also observed. In negative ionization mode, Kaempferol-3-O-glucoside was identified in 447.0925\_3.2505 min  $[M-H]^-$  at level 2. Kaempferol isomers are currently studied for their antioxidant and therapeutic properties, including potential applications in neurodegenerative diseases and cancer [40,41].

The chlorogenic acid quinone Q-CQA was identified at level 2 in negative ionization mode 351.071\_2.5995 min  $[M-H]^-$ . Q-CQS is an oxidative derivative of caffeoylquinic acids formed during sample preparation due to cellular membrane disruption, which can lead to enzymatic browning [42,43].



**Fig. 1.** The bar chart represents the 5-CQA concentration through different factors. (A) Effect of chakra (location) in the 5-CQA concentration. (B) Effect of age-related light factor and age-related shade factor in the 5-CQA concentration. For the age-related light factor and age-related shade factor, age-related light, and age-related shade factor the t-test was used; for the location, post-hoc Tukey was performed to see the significant differences reflected in the ANOVA.

**Table 3**  
Metabolites identified in *Ilex guayusa* leaves by UPLC-MS/MS.

| IL | Metabolite               | RT (min.) | Experimental mass (m/z) [M-H] <sup>-</sup> | Experimental mass (m/z) [M+H] <sup>+</sup> | Formula                                                      | Error (ppm) | MS/MS score |
|----|--------------------------|-----------|--------------------------------------------|--------------------------------------------|--------------------------------------------------------------|-------------|-------------|
| II | Hyperoside               | 3.067     | 463.0874                                   | -                                          | C <sub>21</sub> H <sub>20</sub> O <sub>12</sub>              | -0.54       | 0.7911      |
| II | Trehalose                | 0.583     | 387.1137                                   | -                                          | C <sub>12</sub> H <sub>22</sub> O <sub>11</sub>              | 6.78        | 0.7731      |
| II | Sucrose                  | 0.583     | 341.1082                                   | -                                          | C <sub>12</sub> H <sub>22</sub> O <sub>11</sub>              | -0.54       | 0.7283      |
| II | Lauryl diethanolamide    | 5.637     | -                                          | 288.2529                                   | C <sub>16</sub> H <sub>33</sub> NO <sub>3</sub>              | -3.36       | 0.8484      |
| II | Chlorogenic acid quinone | 2.599     | 351.071                                    | -                                          | C <sub>16</sub> H <sub>16</sub> O <sub>9</sub>               | -1.72       | 0.7882      |
| II | Plantagin                | 3.198     | 447.0926                                   | -                                          | C <sub>21</sub> H <sub>20</sub> O <sub>11</sub>              | -0.30       | 0.8536      |
| II | Acetyl tributyl citrate  | 7.231     | -                                          | 403.2318                                   | C <sub>20</sub> H <sub>34</sub> O <sub>8</sub>               | -3.45       | 0.8057      |
| II | Keracyanine              | 3.017     | 594.1544                                   | -                                          | C <sub>27</sub> H <sub>31</sub> O <sub>15</sub> <sup>+</sup> | -6.85       | 0.7228      |
| II | Kaempferol-3-O-glucoside | 3.250     | 447.0925                                   | -                                          | C <sub>21</sub> H <sub>20</sub> O <sub>11</sub>              | -0.52       | 0.8793      |
| I  | Caffeine                 | 2.423     | -                                          | 195.0885                                   | C <sub>8</sub> H <sub>10</sub> N <sub>4</sub> O <sub>2</sub> | 1.53        | 0.9573      |
| I  | Rutin                    | 2.881     | 609.1454                                   | -                                          | C <sub>27</sub> H <sub>30</sub> O <sub>16</sub>              | -0.26       | 0.8657      |
| I  | Neochlorogenic Acid      | 2.337     | 353.0883                                   | -                                          | C <sub>16</sub> H <sub>18</sub> O <sub>9</sub>               | 2.95        | -           |
| I  | Chlorogenic Acid         | 2.6       | 353.0870                                   | -                                          | C <sub>16</sub> H <sub>18</sub> O <sub>9</sub>               | -0.72       | -           |
| II | Quinic acid              | 0.588     | 191.0555                                   | -                                          | C <sub>7</sub> H <sub>12</sub> O <sub>6</sub>                | -0.33       | 0.8759      |
| II | Valerophenone            | 7.116     | -                                          | 163.1120                                   | C <sub>11</sub> H <sub>14</sub> O                            | -1.77       | 0.8196      |
| I  | Theobromine              | 1.952     | -                                          | 181.0724                                   | C <sub>7</sub> H <sub>8</sub> N <sub>4</sub> O <sub>2</sub>  | -0.56       | -           |

The predominant chlorogenic acid, 5-CQA is associated with multiple biological roles, including fungicidal activity, antioxidant capacity, neuroprotection, modulation of lipid metabolism [44,45]. And anti-inflammatory effects. Q-CQA, as the oxidative form of 5-CQA, can interact with nucleophilic biomolecules such as proteins, lipids, and DNA, potentially affecting cellular redox balance and inducing oxidative

stress [46].

Metabolite profiling employed the FastDDA acquisition method, chosen for its broader analytical coverage compared to conventional DDA [47]. Both FastDDA configurations (5 and 15 precursor ions) identified comparable numbers of metabolites (Supporting information Table S8). Using fewer precursors ions increase the risk of missing

co-eluting compounds with similar mass or retention time, such as caffeoylquinic acid isomers (e.g., 5-CQA 353.0870\_2.6 min [M-H]<sup>-</sup> and 3-CQA 353.0883\_2.3375 min [M-H]<sup>-</sup>), which is crucial in untargeted metabolomics [47].

The FastDDA acquisition of 5 and 15 ions increased MS/MS data acquisition for metabolite identification by spectral comparison, but reduced the MS1 peak shape quality. Similar effects have been described for other data-dependent and data-independent acquisition methods, such as IDA, Swatch, MS<sup>all</sup> [9]). These methods (Swatch and MS<sup>all</sup>) produce a large number of MS/MS spectra but often at the expense of MS1 peak definition. This occurs because selecting a high number of precursor ions for fragmentation reduces the number of MS1 scans available to form the chromatographic peak [48]. Consequently, peak shape quality decreases, and in some cases, peaks may not form correctly (Supplementary information Fig. S7.1). Between the two FastDDA configurations, acquisition of 5 ions yielded higher-quality MS1 peak shapes compared to the 15-ion acquisition. However, this improvement came at the cost of fewer MS/MS spectra per peak, thus limiting metabolite identification.

To mitigate the loss of peak quality, MS/MS data of FastDDA files were separated and subsequently merged with full-scan MS1 data using the "Raw data file merging" module in Mzmine [22]. This integration improved MS1 peak definition in both FastDDA configurations (5 and 15 ions). The resulting datasets were analyzed independently rather than through a direct "merge vs. merge" comparison ([https://github.com/IKIAM-NPLab/I\\_guayusa\\_caffeoylquinic\\_acid\\_isomers/tree/main/Supplementary\\_files](https://github.com/IKIAM-NPLab/I_guayusa_caffeoylquinic_acid_isomers/tree/main/Supplementary_files)).

In negative ion mode, similar metabolites were detected for both FastDDA-5 and FastDDA-15 acquisitions, with consistent trends in positive mode. The main distinction, in both ionization modes, was the significantly higher number of "rows" (MS1 scans associated with MS/MS data) observed in the FastDDA-15 dataset (Supplementary information Fig. S7.2). This increase reflects the larger volume of fragmentation data obtained when targeting 15 precursor ions compared to 5 ions [49].

Comparative analysis of FastDDA datasets, considering both the number of precursor ions (5 ions vs. 15) and the application of raw data file merging, revealed a complex interplay between acquisition parameters and data processing strategies that strongly affects overall data quality. Examination of both positive and negative ionization modes demonstrates a consistent pattern: merged datasets exhibit a markedly higher number of detected features (43% more distinct features) than their non-merged counterparts. This improvement stems from enhanced MS1 peak shape and signal clarity due to the reintegration of full scan data and MS/MS data during merging.

Interestingly, merged datasets acquired with only 5 precursor ions outperformed the non-merged 15-ion datasets, exhibiting 21.35% more distinct features. This highlights the ability of data merging to compensate for reduced fragmentation depth by restoring peak quality. Conversely, merged 15-ion datasets maintained their advantage in total feature count, as broader MS/MS coverage enhances identification [49], provided that peak integrity is preserved through merging.

Overall, the combined use of FastDDA-5 and FastDDA-15 acquisitions with raw data merging represents an analytical strategy. This dual approach maximizes metabolite identification depth while maintaining chromatographic feature fidelity. Collectively, these findings emphasize the effectiveness of integrating acquisition and data processing optimizations to improve compound identification and facilitate propagation in molecular networking analyses.

### 3.4. Molecular networks

The molecular network was constructed following the MZmine workflow, which provides a robust framework for data processing [50]. Using the FBMN (future base molecular networking) approach [26] in combination with GNPS Merge Polarity tool, data from both positive and

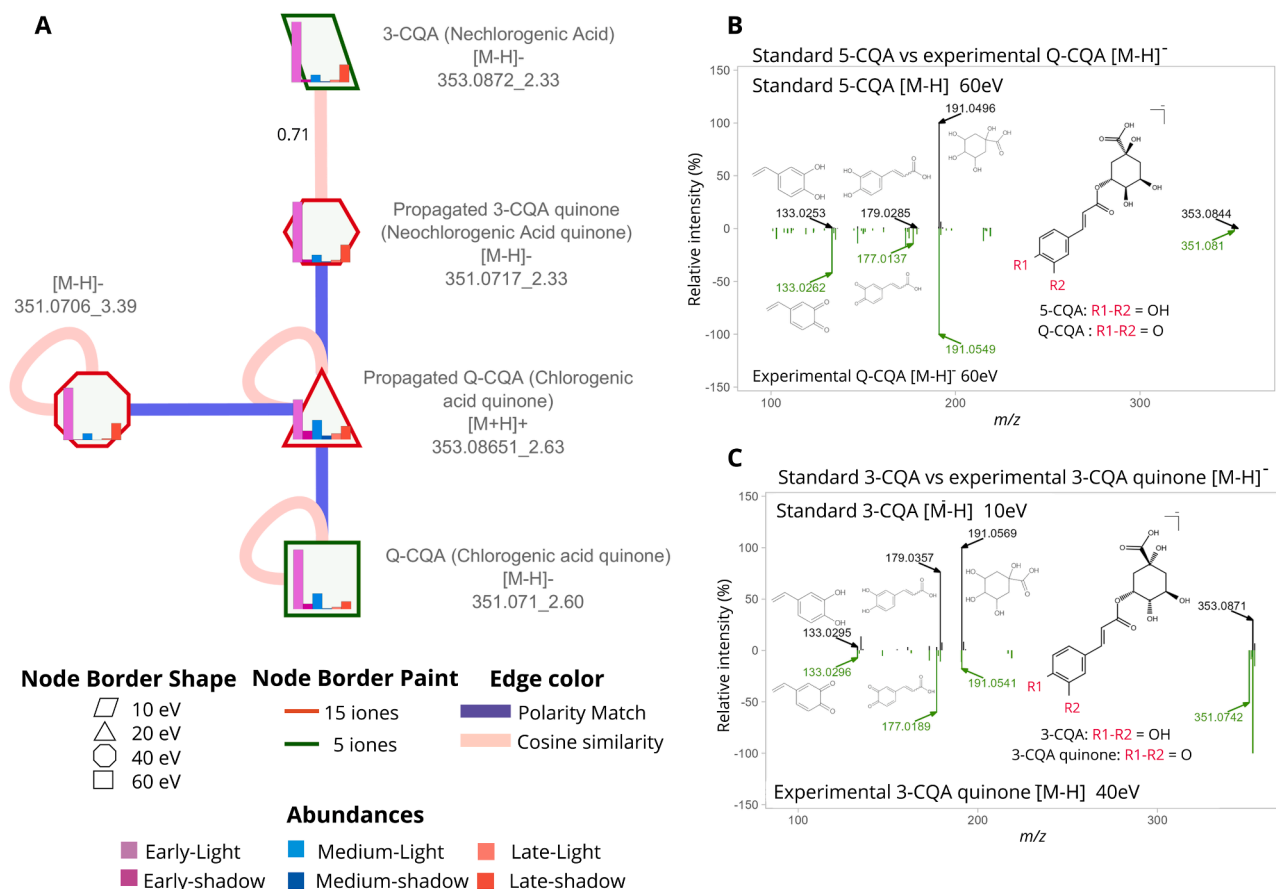
negative ionization modes were integrated to generate an expanded molecular network. This strategy increased the detection of metabolites that were not identified through conventional library-based searches, thereby enhancing the biochemical characterization and research value of *Ilex guayusa*.

To fully elucidate the structure of the molecular network structure, it is essential to characterize the oxidative specificities associated with the diverse isomeric variants of caffeoylquinic acids [43]. These compounds consist of a caffeic acid moiety esterified to a quinic acid backbone [42]. The oxidative process, typically catalyzed by enzymatic mechanisms (polyphenol oxidase enzyme), occurs mainly on the aromatic ring of the caffeic acid unit. This reaction converts chlorogenic acid into its oxidized derivative, chlorogenic acid quinone [42]. Such oxidative behavior is considered a general feature among all caffeoylquinic isomers [51–54]. Accordingly, each isomer can exist in two fundamental configurational states: the non-oxidized form (e.g., chlorogenic acid) and its oxidized counterpart (chlorogenic acid quinone). The visualization of fragmentation spectral profiles through mirror plots supports these oxidative transformations. Fig. 2B presents a mirror plot of comparison between the chlorogenic acid (5-CQA) standard and its oxidized form chlorogenic acid quinone (Q-CQA). Their precursor ion are (*m/z* rt) 353.0844\_2.61 min [M-H]<sup>-</sup> and 351.081\_2.59 min [M-H]<sup>-</sup>, respectively, differing by approximately 2 Da and 0.02 min. Fragment analysis confirms the oxidation mechanism: the quinic acid fragment (191.0496 Da) remains unchanged in both spectra, while the caffeic acid fragment shift from 179.0285 Da in 5-CQA to 177.0137 Da, consistent with a 2 Da mass difference attributed to oxidation. This mass shift accounts for both the difference in ions mass and the observed discrepancy in fragment ion *m/z* values. Additionally, 5-CQA exhibits a characteristic caffeic acid-derived fragment at 133.0253 Da, while Q-CQA shows a similar fragment at 133.0262 Da, corresponding to oxidized caffeic acid. The difference between these fragments lies in the aromatic ring structure, where Q-CQA loses a double bond.

Neochlorogenic acid (3-CQA) was identified at level 1 confidence based on accurate mass, retention time and fragmentation pattern (Supplementary information Fig. S6). The node identified as 3-CQA with 353.0872\_2.33 min [M-H]<sup>-</sup> shows an interconnection with an adjacent node at 351.0717\_2.33 min [M-H]<sup>-</sup>, which exhibits a molecular mass similar to that of Q-CQA 351.0710 Da but elutes at different chromatographic retention time (2.33 min and 2.60 min respectively). Comparative analysis of their MS/MS fragmentation pattern within the molecular network showed that the 3-CQA node is connected to the 351.0717\_2.33 min [M-H]<sup>-</sup> node with a cosine similarity of 0.71 (Fig. 2A). The mirror plots (Fig. 2C) revealed a fragmentation pattern similar to that in Fig. 2B. Based on their structural similarity and characteristic isomerism, we propose that this node represents the oxidized form of neochlorogenic acid and therefore annotate it as neochlorogenic acid quinone (3-CQA quinone).

Within molecular network, polarity-based interconnections revealed a node detected in positive mode (353.08651\_2.63 min [M+H]<sup>+</sup>) linking the propagated 3-CQA quinone node (351.0717\_2.33 min [M-H]<sup>-</sup>) and the unknown node at 351.0706\_3.39 min [M-H]<sup>-</sup>. These two latter nodes exhibit a substantial difference in retention time ( $\geq 0.05$  min), which likely results from the wide time-tolerance parameters configured within GNPS analytical criteria. Additionally, the unknown node in positive ionization mode is connected with the Q-CQA node (351.071\_2.60 min [M-H]<sup>-</sup>), showing a close retention time ( $\Delta t = 0.03$  min) and a mass difference of 2.01551 Da, consistent with ionization mode. By examining the extracted ion chromatograms (EIC), *m/z* ratios and retention times of these nodes, we propose that this feature corresponds to the positive-ionized form of Q-CQA (Supplementary information Fig. S8).

As shown in Fig. 2A, where relative feature abundances are represented by bar plots within the nodes, these abundance profiles agree with the quantitative results for 5-CQA (Fig. 1D). Specifically, caffeoylquinic acid isomers are more abundant at early ages under sunlight



**Fig. 2.** The molecular network of caffeoylquinic acid isomers shows node-to-node junctions where: (A) CQA molecular network with node shape encoding collision energy (parallelogram, 10 eV; triangle, 20 eV; octagon, 40 eV; square, 60 eV). Node edge color denotes FastDDA acquisition method (red, 15 ions; blue, 5 ions). Edges are colored pink when spectral cosine similarity  $\geq 0.70$  and blue when linked by retention-time alignment indicating a polarity match. (B) Mirror plot comparing the experimental spectrum annotated as Q-CQA with a public library spectrum of 5-CQA. (C) Mirror plot comparing the spectrum of the node propagated as 3-CQA quinone with a reference spectrum of 3-CQA.

conditions. As plants age, changes in sunlight exposure alter the relative abundance of these isomers, with shaded plants showing higher late-stage accumulation compared to those grown under sunlight exposure. It is also observed that at the node border, all propagated features corresponded to the FastDDA 15 ions acquisition. These features showed lower precursor abundances and were therefore not detected under the FastDDA with 5 ions. These findings support the use of both acquisition methods (FastDDA with 5 and 15 ions) to comprehensively capture the chemical diversity of plant metabolome.

### 3.5. Multivariate analysis

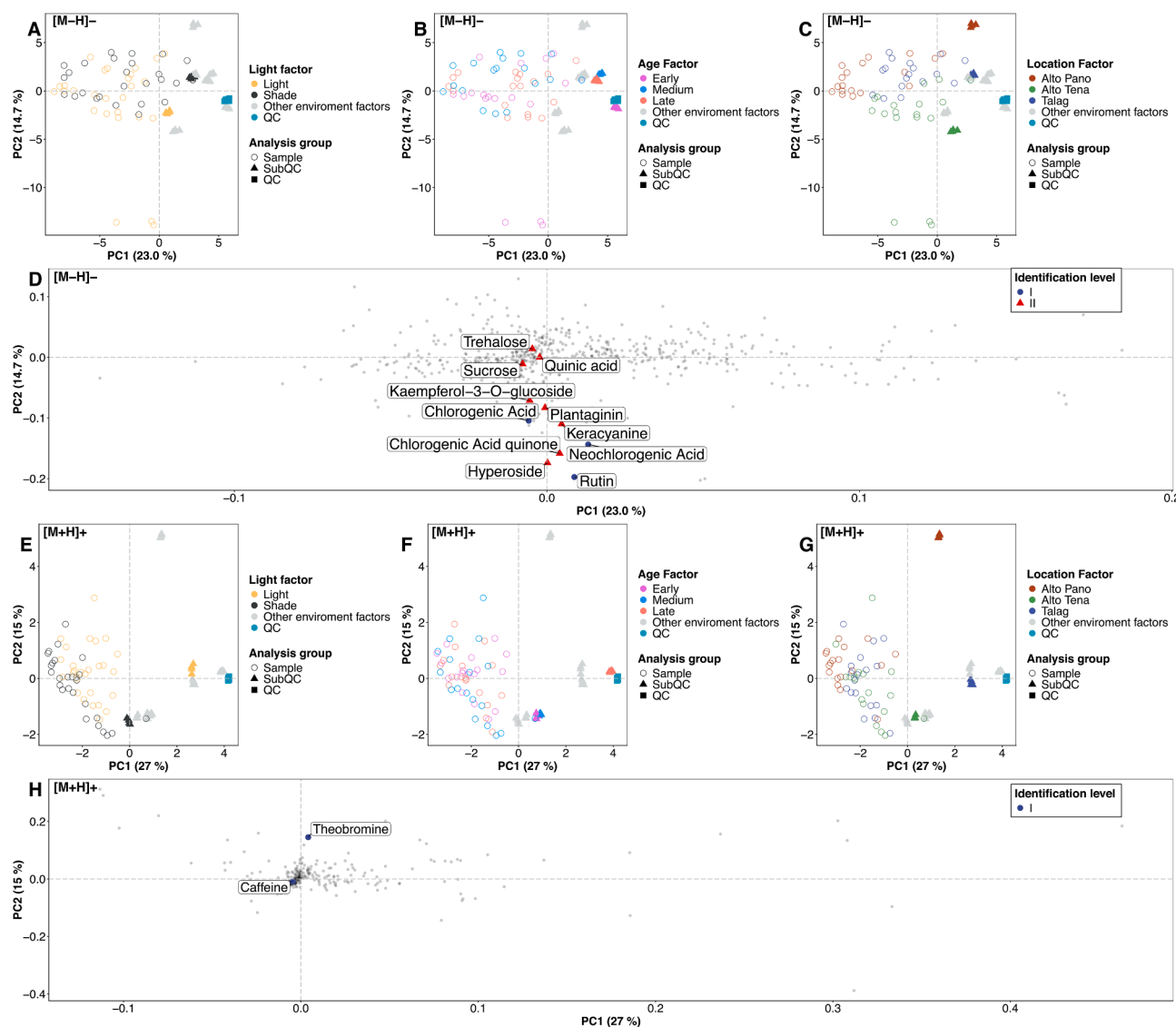
Following metabolite identification and molecular network analysis, multivariate statistical analyses were performed, including principal component analysis (PCA), volcano plot and heatmap with hierarchical cluster analysis (HCA). The methodologies employed for these analyses, along with detailed data processing steps, are available in the GitHub repository ([https://github.com/IKIAM-NPLab/I\\_guayusa\\_caffeoylquinic\\_acid\\_isomers](https://github.com/IKIAM-NPLab/I_guayusa_caffeoylquinic_acid_isomers)).

The PCA score plots for both ionization modes (Fig. 3) reveal a clear separation of samples into three primary analytical groups: “Sample” (individual leaf extracts collected under varying conditions), “SubQC” (pooled samples from each experimental condition, used to assess factor-specific variations), and “QC” (a pooled mixture of all samples). QC samples were essential for monitoring system stability and ensuring optimal UPLC-MS/MS performance throughout the analytical sequence. Each experimental factor is highlighted in a separate PCA score plot to

illustrate its specific influence on the metabolomic profile of *Ilex guayusa* leaves (Fig. 3A - 3C)).

The corresponding PCA loading plots (Fig. 3D and 3H) were subsequently used to identify the key metabolites contributing most to sample differentiation across the tested conditions.

In general, the PCA score plot corresponding to the “samples” groups did not exhibit a clear separation of light and age factors along the two principal components (PC1 and PC2) under either of the two analytical ionization modes (positive and negative) (Fig. 3A-B and Fig. 3E-F). This limited separation contrasts with the SubQC projections, which displayed a markedly clearer grouping across all three evaluated factors (light, age, and location), reinforcing that the underlying multivariate signal distinguishing the groups is present but becomes partially obscured in individual samples due to biological and technical variability. Nevertheless, a notable exception within the samples was the location factor, for which consistent clustering patterns were observed (Fig. 3C and Fig. 3G); Under negative ionization mode (Fig. 3C) two clusters were observed along the negative first principal component (PC1): one cluster grouped most Alto Pano samples, while another cluster, located closer to the origin grouped the Alto Tena and Talag samples; on the other hand, along the second principal component (PC2), most Alto Tena samples were clustered in the negative PC2, whereas the majority of Alto Pano and Talag samples were clustered in the positive PC2. Similarly, under positive ionization mode (Fig. 3G), samples were mainly grouped according to location, with most Alto Pano samples separated along PC1 from the cluster formed by Alto Tena and Talag. These patterns indicate that the location factor exerted a



**Fig. 3.** Principal component analysis (PCA) of the deconvoluted MS1 feature table from different experimental factors in *Ilex guayusa* leaves in positive and negative ionization modes. (A) PCA score plot in negative ionization mode highlighting the light exposure factor, (B) plant in different ages, and (C) geographical location, (D) the corresponding PCA loading plot of MS1 features; for positive ionization mode (E) light exposure, (F) plant in different ages, (G) geographical location, (H) and the corresponding PCA loading plot of MS1 features.

stronger influence on the metabolomic profile of *Ilex guayusa* than light exposure or plant age, since the latter factors did not produce a clear separation among the samples in both ionization modes.

According to the PCA loadings plots, the edaphic and climatic conditions at Alto Tena appear to promote the accumulation of caffeoyl-quinic acid isomers, which contribute to the separations of Alto Tena samples located in the negative PC2 (Fig. 3D). In contrast, the edaphic conditions at Alto Pano favor a higher accumulation of theobromine, reflecting local adaptations to abiotic stress (Fig. 3H). This pattern may result from the relative chemical homogeneity of soils or from localized environmental effects that modulate the availability of essential nutrients, such as nitrogen, phosphorus, and potassium, as discussed by Fernández et al. (2025) [13]. Although methylxanthines such as caffeine and theobromine appear in the PCA loadings as abundant constituents of *Ilex guayusa*, they were included solely as part of the untargeted metabolomic profile. These compounds were not subjected to targeted quantification or method validation, as the analytical workflow developed in this study was specifically optimized for chlorogenic acids. Nevertheless, the presence and relative abundance of these metabolites, together with the observed variation in chlorogenic acid isomers, can be

interpreted within a broader environmental context. In studies of medicinal plants, variations in soil fertility (particularly N and P) have been shown to correlate directly with differences in the accumulation of secondary metabolites [55]. Furthermore, nutrient availability can significantly influence phenolic metabolism, including pathways involved in the biosynthesis of chlorogenic acid isomers. Variations in soil nutrient content may modulate the accumulation of these compounds, either directly or indirectly, by inducing physiological responses such as oxidative stress under nutrient-deficient conditions [56]. Therefore, the grouping according to location level detected in the PCA reflects, at least in part, differences in soil fertility among Alto Pano, Alto Tena and Talag, which determine the differential biosynthesis of compounds in *Ilex guayusa* leaves.

However, the clear separation of the samples by location in the global PCA reflects the dominant influence of this factor as the primary discriminant. To identify the subsequent discriminating factors, we performed location specific PCA analyses, thereby removing the influence of location and allowing the remaining factors to separate according to their intrinsic multivariate structure (Supplementary Figs. S9–S10). Through this approach, sunlight exposure consistently

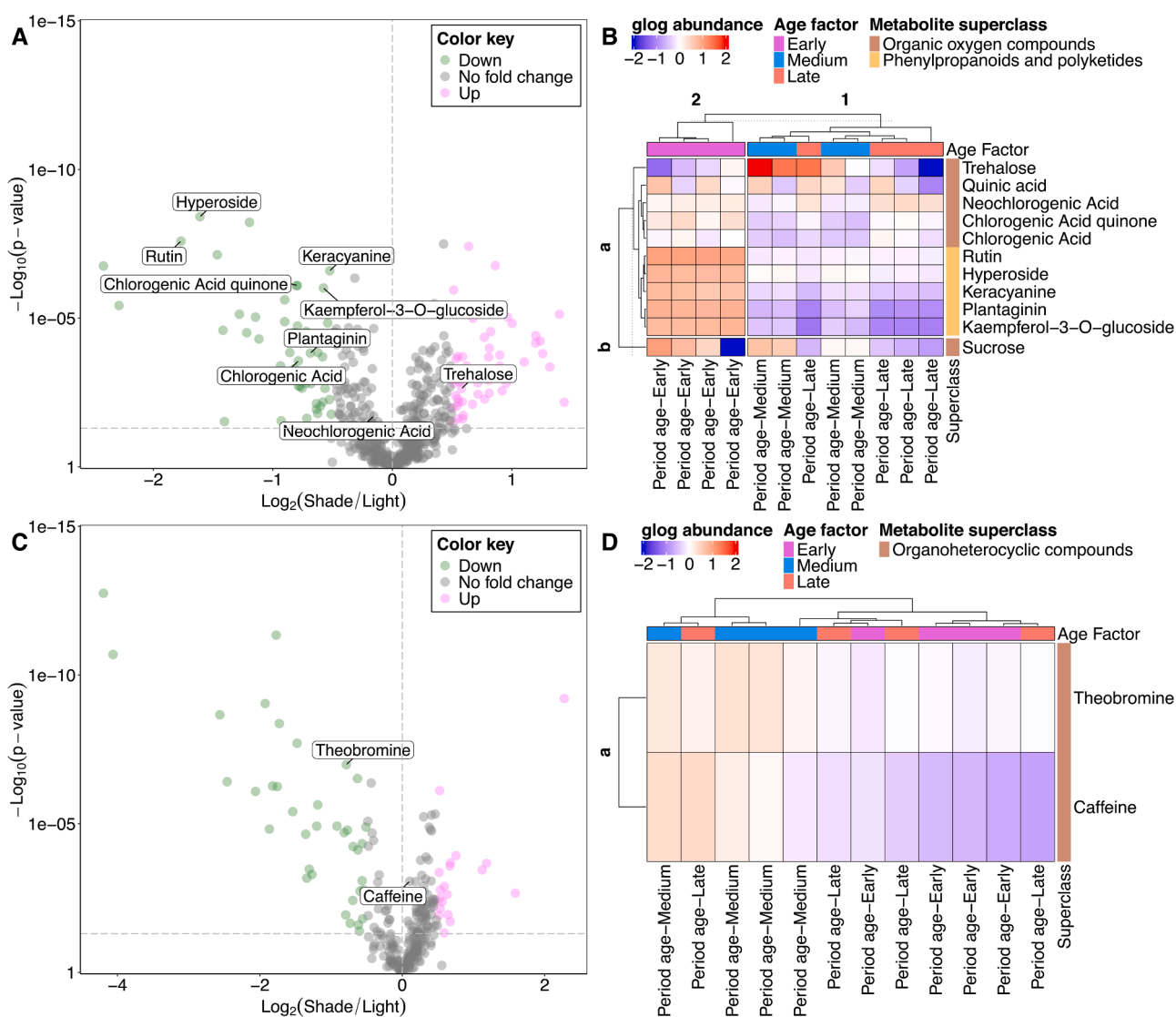
emerged as the second discriminating factor, followed by plant age as the third, demonstrating that these effects are intrinsically present once the dominant spatial variation is accounted.

To further support these results and enhance interpretability in the presence of biological variability observed in the global PCA, a SubQC (Subgroup Quality Control) approach [57] was implemented, generating composite profiles for each level of light and age. These SubQC samples reduced biological noise and yielded markedly clearer group separations, enabling a more robust evaluation of the metabolomic shifts associated with sunlight exposure and plant age. Building upon this improved discrimination, differential metabolite abundance for the light factor was assessed using volcano plot, and age related metabolomic patterns were examined using heatmap HCA analyses in both ionization modes.

As explained above for the light factor, a volcano plot was generated for both ionization modes (Fig. 4A for negative ionization mode and Fig. 4C for positive ionization mode). In these plots, a color key facilitates the categorization of each data point, where each dot represents an

individual metabolite: green dots indicate significantly down-regulated metabolites (i.e., lower relative abundance under shade compared with light conditions); pink dots indicate significantly up-regulated metabolites (i.e., higher relative abundance under shade compared with light conditions); and gray dots correspond to metabolites with no significant fold change or those not meeting statistical significance threshold ( $p < 0.05$ ). In negative ionization mode (Fig. 4A), several metabolites such as caffeoylquinic acid isomers exhibited statistically significant differences, showing higher concentration ( $\log_2$  fold change  $< 0.05$ ) under light conditions than under shade, consistent with previous reports [58]. In contrast, under positive ionization mode (Fig. 4C), caffeine did not display a notable fold change, although its variation was statistically significant ( $p < 0.05$ ) under shade conditions. Conversely, theobromine exhibited a significant fold change ( $\log_2$  fold change  $< 0.05$ ) under light conditions, which aligns with prior studies describing the regulation of methylxanthines within the *Ilex* genus [13].

To interpret the influence of the age factor, a heatmap integrated with hierarchical cluster analysis (HCA) (Fig. 4B and Fig. 4D), was used



**Fig. 4.** This figure comprises two distinct panels: panel (A) presents a volcano plot in negative ionization that depicts the magnitude of fold changes and statistic significance in metabolite concentrations attributable to differential solar exposure, while panel (B) presents a heatmap integrated with hierarchical cluster analysis (HCA) in negative ionization to illustrate variations in metabolite abundance related with age factors. In positive ionization the panel (C) presents a volcano plot that depicts the magnitude of fold changes and statistical significance in metabolite concentrations attributable to differential solar exposure, while (D) presents a heatmap in positive ionization integrated with hierarchical cluster analysis (HCA) to illustrate variations in metabolite abundance related with age factors.

to visualize the relative abundance profiles of identified metabolites across distinct sample groups, categorized by age factor (early, medium, and late). In this representation, each row corresponds to a metabolite, while columns represent subQC data systematically grouped by age level. The color scale denotes normalized metabolite abundance, with blue shades indicating lower relative abundance and warmer colors (red) indicating high relative abundance. The metabolite dendrogram clusters compounds based on the similarity of their abundance profiles across samples.

In the negative ionization mode the HCA (Fig. 4B) revealed two main clusters: one predominantly comprising samples from the medium and late age groups, generally characterized by lower relative abundance of most metabolites; and another composed mainly of early-age samples, which exhibited higher relative abundance of caffeoylquinic acid isomers, likely associated with initial defense mechanisms in plant's secondary metabolism [59]. Under positive ionization mode (Fig. 4D), the medium-age group was more closely associated with higher abundances of caffeine and theobromine. Multivariate analyses indicate that the *Ilex guayusa* metabolome is driven primarily by location edaphic conditions, with light and age as secondary modulators. This pattern reflects adaptive metabolic plasticity. Integrating PCA, volcano plot, and HCA clarifies the factor hierarchy and motivates functional studies linking biosynthesis to ecological resilience and phytochemical diversity.

#### 4. Conclusions

This study establishes a comprehensive analytical workflow that integrates validated HPLC–UV–vis quantification with optimized FastDDA-based UPLC–MS/MS acquisition, raw-data merging, molecular networking, and multivariate chemometric analysis for the study of chlorogenic acids in plant-derived samples. Robust and reproducible quantification was achieved for the major chlorogenic acid (5-CQA) using HPLC–UV, while the LC–MS/MS strategy was specifically optimized to improve MS1 peak integrity, expand MS/MS coverage, and increase confidence in structural annotation, rather than to perform full isomer-resolved MS quantification.

The combined FastDDA and raw-data merging approach significantly enhanced data quality, resulting in improved chromatographic peak definition and a higher number of detected features, which facilitated the reliable annotation and propagation of structurally related caffeoylquinic acid isomers and their oxidized derivatives within feature-based molecular networks. The integration of molecular networking with multivariate analyses (PCA, volcano plots, and HCA) enabled the interpretation of environmentally driven metabolic variation, revealing that geographic location exerted the strongest influence on the metabolomic profile, followed by sunlight exposure and plant age.

Overall, the proposed workflow provides a transferable and scalable analytical framework for the structural characterization and interpretation of chlorogenic-acid-related chemistry across complex plant and natural product matrices. While it does not aim to deliver complete isomer-specific quantitative resolution by MS, it offers a practical methodological reference that bridges validated chromatographic quantification with advanced MS acquisition and data-processing strategies, supporting high-confidence metabolite annotation in natural products and food chemistry research.

#### Data availability

Data and code are openly available in a public repository: [https://github.com/IKIAM-NPLab/I\\_guayusa\\_caffeoylquinic\\_acid\\_isomers](https://github.com/IKIAM-NPLab/I_guayusa_caffeoylquinic_acid_isomers).

#### CRedit authorship contribution statement

**Thomas Garzón:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis. **Jefferson V. Pastuñafasso:** Methodology, Data curation. **Cristian Quiroz-Moreno:** Methodology, Data curation. **Melanie Ochoa-Ocampo:** Writing – review & editing, Methodology. **Evencio J. Medina-Villamizar:** Visualization, Validation, Supervision, Methodology. **Nina Espinosa de los Monteros-Silva:** Project administration, Methodology. **Karel Diéguez-Santana:** Writing – review & editing, Supervision, Conceptualization. **Jessica L. Cooperstone:** Writing – review & editing, Methodology. **Zulay Niño-Ruiz:** Supervision, Resources, Project administration, Funding acquisition. **José R. Almeida:** Writing – review & editing, Data curation. **Noroska G.S. Mogollón:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.chroma.2026.466732](https://doi.org/10.1016/j.chroma.2026.466732).

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